The algebraicity threshold for orthogonal tensor multiplicities

Fixed representation types and symmetric determinantal Hilbert series

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Abstract

Let V be the defining representation of the complex orthogonal group O(m), and let W be any fixed nonzero finite-dimensional algebraic representation. We give a complete dimension threshold for the ordinary generating function of the multiplicities of W in tensor powers of V: it is always D-finite and is algebraic exactly for m at most three. Explicit formulas cover every irreducible type in the algebraic range. In higher dimensions, character-weighted trace asymptotics and the arithmetic of the leading constant exclude algebraicity on each compatible central parity. The scalar specialization gives the threshold for reduced symmetric determinantal quotients in the symmetric-sequence convention. The scalar asymptotics are due to Regev; the rank-three invariant radical is a specialization of Almkvist–Dicks–Formanek; the D-finiteness input and algebraic-function obstruction are established tools. The contribution presented here is the assembled all-dimension, all-fixed-type classification and its no-restoration consequence, not new asymptotics or a claim of established historical priority. Exact finite diagnostics accompany the written proof.

1. Statement and conventions

Let the ground field have characteristic zero. Put

$$B_m = \mathrm{Sym}(\mathrm{Sym}^2(\mathbf{C}^\infty))/I_{m+1}, \qquad m \geq 0,$$

where I_{m+1} is the ideal generated by the $(m+1)$-minors of the generic symmetric matrix. Thus m denotes the allowed rank, not the size of a vanishing minor. Write f^μ for the dimension of the Specht representation of shape μ, and $2\lambda = (2\lambda_1, 2\lambda_2, \ldots)$.

The non-exponential Hilbert series in the symmetric-sequence convention is

$$H_m(t) = \sum_{n\geq 0} a_m(n) t^{2n}, \qquad a_m(n) = \sum_{\substack{\lambda \vdash n \\ \ell(\lambda)\leq m}} f^{2\lambda}. \tag{1}$$

In particular, $a_m(0) = 1$. We also use $F_m(z) = \sum_{n\geq 0} a_m(n) z^n$, so $H_m(t) = F_m(t^2)$. Neither series is the ordinary Hilbert series after evaluating the functor on a fixed finite-dimensional vector space; that different series is rational.

Theorem. For every finite integer $m \geq 0$, H_m and F_m are D-finite. They are algebraic over their rational function fields **if and only if** $m \leq 3$. More explicitly,

$$\begin{aligned} H_0(t) &= 1, \\ H_1(t) &= \frac{1}{1-t^2}, \\ H_2(t) &= \frac{1}{2}\left(1 + \frac{1}{\sqrt{1-4t^2}}\right), \\ H_3(t) &= \frac{\sqrt{(1+3t)/(1-t)} - \sqrt{(1-3t)/(1+t)}}{4t}. \end{aligned} \tag{2}$$

The square roots in (2) have constant term 1; the apparent singularity of H_3 at zero is removable. For every fixed $m \geq 2$,

$$a_m(n) \sim c_m m^{2n} n^{-\alpha_m}, \qquad \alpha_m = \frac{m(m-1)}{4}, \tag{3}$$

where

$$c_m = 2^{1-m}\left(\frac{m}{2}\right)^{\alpha_m} \pi^{-m/2} \prod_{j=1}^{m} \Gamma(j/2) > 0. \tag{4}$$

This is one formula valid for every fixed rank. No uniform error estimate in a regime where m grows with n is asserted.

The theorem addresses the reduced determinantal-quotient formulation preserved in the local UnsolvedMath record AIM-REPRESENTATION_THEORY-0022. The catalogue excerpt and provenance are included; the live AIM source has not been independently recovered. It does not assert a classification for arbitrary modules or nonreduced determinantal thickenings. Sections 9–12 prove the same threshold for every fixed nonzero finite-dimensional orthogonal representation. D-finiteness, the invariant-theory identification, coefficient asymptotics, and nearby examples have substantial antecedents. No historical priority follows merely from the proofs supplied here; see Section 14 and `CITATION_AUDIT.md`.

Conceptually, the question asks when symmetry-constrained tensor counts admit an algebraic generating function. The geometric dimension of the orthogonal group fixes the polynomial decay of the counts. In one residue class that decay forces a logarithm; in the other, the exact leading constant rules out an algebraic square-root singularity. Both the exponent and the constant matter: a half-integer exponent by itself would not prove transcendence.

2. The invariant-theory bridge

We spell out the identification so that the grading and the use of the full orthogonal group are fixed before the analytic argument. In this section take $m \geq 1$; the rank-zero quotient is the ground field.

Let $V = \mathbf{C}^m$ have its standard nondegenerate symmetric form. The first and second fundamental theorems for the orthogonal group say that polynomial invariants of finitely many vectors are generated by their pairwise inner products, with relations generated by the $(m+1)$-minors of their Gram matrix. Equivalently, naturally in the auxiliary vector space U,

$$B_m(U) \cong \mathrm{Sym}(U \otimes V)^{O(V)}. \tag{5}$$

Duals can be inserted if one uses coordinate rings instead of covariant polynomial functors; they do not affect the following multiplicities. The invariant form identifies V with its dual. A product of n Gram entries has tensor degree $2n$ in (5).

For clarity, the multilinear version of the two fundamental theorems has a particularly direct interpretation here. The degree-$2n$ component of the free symmetric tca is spanned by perfect matchings of $2n$ labelled positions. A matching maps to the product of contractions over its pairs. These contractions span the invariant tensors. Every linear relation among them is a linear combination of a determinant of $m+1$ contractions, multiplied by a matching on the remaining positions. Those are exactly the multilinear components of I_{m+1}. These statements are the classical fundamental theorems in the formulations recalled as Theorems 3.1 and 3.4 of Lehrer–Zhang [LZ]. Thus (5) identifies the quotient, not merely a subspace of equal dimension.

In the symmetric-sequence model of polynomial functors, $\mathrm{Sym}(U \otimes V)$ corresponds to the sequence $V^{\otimes d}$, with its place-permutation action of S_d. Consequently (5) gives

$$(B_m)_d \cong (V^{\otimes d})^{O(V)}. \tag{6}$$

One can also check (1) directly. Schur–Weyl decomposition gives

$$V^{\otimes d} \cong \bigoplus_{\substack{\mu \vdash d \\ \ell(\mu) \le m}} S_\mu(V) \otimes [\mu].$$

The orthogonal invariant multiplicity in $S_\mu(V)$ is one when every row of μ is even, and zero otherwise. This standard restriction identity is stated explicitly at the start of Section 3 of Rains [R]. Taking invariant dimensions therefore gives exactly (1). It also proves that the quotient is bounded: every Schur functor occurring has at most m rows.

The real compact group $K = O(m, \mathbf{R})$ has the same invariant tensors as its complexification $O(m, \mathbf{C})$. For example, invariance under the real Lie algebra implies invariance under its complexification and hence under the complex identity component; a real reflection accounts for the other component. Averaging over normalized Haar measure on K is the projection onto the invariant subspace. Its trace on $V^{\otimes d}$ is therefore

$$\dim(V^{\otimes d})^{O(V)} = \int_K (\operatorname{tr} g)^d \, dg. \tag{7}$$

Since multiplication by $-I$ preserves Haar measure, the odd moments vanish. Equations (1), (6), and (7) prove

$$a_m(n) = \int_K (\operatorname{tr} g)^{2n} \, dg, \qquad H_m(t) = \int_K \frac{dg}{1 - t \operatorname{tr} g}, \quad |t| < 1/m. \tag{8}$$

The interchange of summation and integration is justified by a uniformly convergent geometric series, since $|\operatorname{tr} g| \le m$.

All algebraic objects and their identifications can be formed over $\mathbf{Q}$. Extension of scalars preserves the finite-dimensional graded dimensions. Hence using complex and compact groups to prove (8) does not restrict the characteristic-zero conclusion.

3. Exact group volume and coefficient asymptotics

The scalar asymptotic (3)–(4) is already a direct specialization of Regev [Re, Definition 3.1, Theorem 3.3 and formula (F.4.5.1)]. We include a compact-group rederivation to fix normalization and support

the character-weighted argument in Section 10; neither the scalar asymptotic nor its constant is claimed as new.

3.1. Volume normalization

Equip K with the bi-invariant metric whose value on skew-symmetric matrices is

$$\langle A, B\rangle = -\tfrac{1}{2}\operatorname{tr}(AB). \tag{9}$$

If $A = \sum_{i<j} x_{ij}(E_{ij} - E_{ji})$, then $\|A\|^2 = \sum_{i<j} x_{ij}^2$. Let $D = m(m-1)/2$ and let V_m be the total Riemannian volume of the full, generally disconnected group.

The map $O(m) \longrightarrow S^{m-1}$ sending a matrix to its last column is a Riemannian submersion for (9) and the unit-round-sphere metric. Indeed, at the identity the vertical tangent space is $\mathfrak{o}(m-1)$, and the horizontal vector with entries x_{im} maps to $\sum_{i<m} x_{im}e_i$ with exactly the same norm. Invariance gives the same assertion at every point. Every fiber is isometric to $O(m-1)$. Integrating the Riemannian volume along fibers therefore gives

$$V_m = \operatorname{vol}(S^{m-1})V_{m-1}.$$

Starting with $V_1 = 2$ and using the unit-sphere volume formula yields

$$V_m = \prod_{j=1}^{m} \frac{2\pi^{j/2}}{\Gamma(j/2)} = \frac{2^m \pi^{m(m+1)/4}}{\prod_{j=1}^{m}\Gamma(j/2)}. \tag{10}$$

The sphere formula follows, with this normalization, by evaluating the Gaussian integral on $\mathbf{R}^j$ in polar coordinates. Thus (10) needs no convention-dependent group-volume citation. Normalized Haar measure is the Riemannian volume measure divided by V_m.

3.2. The two maxima

For an orthogonal matrix, equality $|\operatorname{tr} g| = m$ occurs only at I or $-I$. One direct check is

$$\|g - I\|_F^2 = 2m - 2\operatorname{tr} g, \qquad \|g + I\|_F^2 = 2m + 2\operatorname{tr} g.$$

Outside fixed disjoint small neighborhoods of these two points, compactness gives $|\operatorname{tr} g| \le m - \delta$ for some $\delta > 0$.

Near I, use exponential coordinates $g = \exp A$. Odd powers of a real skew-symmetric matrix have trace zero, so

$$\operatorname{tr}(\exp A) = m - \|A\|^2 + O(\|A\|^4).$$

Consequently, in a sufficiently small fixed coordinate ball,

$$\left(\frac{\operatorname{tr}(\exp A)}{m}\right)^{2n} = \exp\left(-\frac{2n}{m}\|A\|^2 + O(n\|A\|^4)\right). \tag{11}$$

The Riemannian volume density in these coordinates is a smooth function $J(A)$ with $J(0) = 1$, because the differential of the exponential map at zero is the identity on the orthonormal coordinate space (9).

Set $A = n^{-1/2}X$. The scaled integrand tends pointwise to $\exp(-2\|X\|^2/m)$. Shrinking the coordinate ball if needed, $2\log(\mathrm{tr}(\exp A)/m) \le -\|A\|^2/m$. Multiplying this inequality by n shows that the scaled integrand, times the bounded density, is bounded by a constant times $\exp(-\|X\|^2/m)$ on its expanding domain, and zero outside that domain. Dominated convergence proves that the contribution from the neighborhood of I is

$$\frac{m^{2n}n^{-D/2}}{V_m}\int_{\mathbf{R}^D} e^{-2\|X\|^2/m}\,dX\,(1+o(1)) = \frac{m^{2n}n^{-D/2}}{V_m}\left(\frac{\pi m}{2}\right)^{D/2}(1+o(1)).$$

Multiplication by $-I$ gives an identical contribution from the other neighborhood. The complement contributes at most $(m-\delta)^{2n}$, which is exponentially smaller. Thus

$$a_m(n) \sim \frac{2}{V_m}\left(\frac{\pi m}{2}\right)^{D/2} m^{2n}n^{-D/2}. \tag{12}$$

Substitution of (10) gives (3)–(4), since $D/2 = \alpha_m$. This also proves that the radius of convergence of F_m is exactly $\rho_m = m^{-2}$.

The factor 2 in (12) counts the two maxima. The volume V_m also includes both connected components. For odd m the maxima are in different components; for even m both are in the identity component. The argument covers both situations without changing normalization.

3.3. The arithmetic part of the constant

Put $r = \lfloor m/2 \rfloor$. The identities

$$\Gamma(j) = (j-1)!, \qquad \Gamma(j+\tfrac{1}{2}) = \frac{(2j)!}{4^j j!}\sqrt{\pi}$$

show from (4) that

$$c_m = A_m \pi^{-r/2}, \qquad A_m \in \overline{\mathbf{Q}}_{>0}. \tag{13}$$

For an entirely rational specification of A_m, define

$$b_{2j} = (j-1)!\ (j \ge 1), \quad b_{2j+1} = \frac{(2j)!}{4^j j!}\ (j \ge 0).$$

Then

$$A_m^2 = 2^{2-2m}\left(\frac{m}{2}\right)^{m(m-1)/2}\prod_{j=1}^{m} b_j^2 \in \mathbf{Q}_{>0}. \tag{14}$$

For example, $c_2 = 1/(2\sqrt{\pi})$, $c_3 = 3\sqrt{6}/(32\sqrt{\pi})$, and $c_4 = 1/(2\pi)$. These provide normalization checks against the explicit low-rank sequences.

4. A radial obstruction to algebraicity

The following elementary consequence of Newton–Puiseux is convenient because it uses only limits along the positive real axis.

Lemma. Let $F(z) = \sum_{n\ge 0} u_n z^n \in \mathbf{Q}[[z]]$, and suppose that for a positive algebraic number ρ,

$$u_n \rho^n \sim c n^{-\alpha}, \qquad c > 0.$$

(a) If α is a positive integer, F is transcendental.

(b) If $\alpha = p + 1/2$ with an integer $p \geq 0$, algebraicity of F implies $c\sqrt{\pi} \in \overline{\mathbf{Q}}$.

Proof. Write $(n)_k = n(n-1)\cdots(n-k+1)$ and, for $0 < x < 1$, put

$$G_k(x) = \rho^k F^{(k)}(\rho x) = x^{-k} \sum_{n \geq k} (n)_k u_n \rho^n x^n. \tag{15}$$

In (a), take $k = \alpha - 1$. The summand coefficient is asymptotic to c/n. Comparison of the tails with $(c \pm \varepsilon)/n$, followed by $\varepsilon \downarrow 0$, gives

$$G_k(x) \sim c \log \frac{1}{1-x}. \tag{16}$$

Finite initial terms do not affect this divergent asymptotic, and $x^{-k} \to 1$.

If F were algebraic, then so would be G_k. An algebraic function has a convergent Laurent–Puiseux expansion at the algebraic point $x = 1$, with rational exponents, algebraic coefficients, and only finitely many negative exponents. Such a branch either has a finite limit or diverges like a nonzero constant times $(1-x)^{-q}$ for some positive rational q. Neither alternative is (16).

In (b), take $k = p$. The summand coefficient in (15) is asymptotic to $cn^{-1/2}$. The elementary Abelian limit is

$$\sum_{n \geq 1} n^{-1/2} x^n \sim \sqrt{\pi}(1-x)^{-1/2}. \tag{17}$$

To verify the constant, put $x = e^{-s}$. Then $s^{1/2} \sum n^{-1/2} e^{-sn}$ is the Riemann sum $s \sum (sn)^{-1/2} e^{-sn}$, tending to $\int_0^\infty y^{-1/2} e^{-y}\, dy = \sqrt{\pi}$. The integrable singularity at zero is controlled by $s^{1/2} \sum_{n \leq \varepsilon/s} n^{-1/2} \leq 2\sqrt{\varepsilon}$, and the tail at infinity is exponentially small. Finally $s \sim 1 - x$. Tail comparison as in (a) now proves

$$G_p(x) \sim c\sqrt{\pi}(1-x)^{-1/2}. \tag{18}$$

If G_p is algebraic, its leading Puiseux exponent must be $-1/2$ and its leading coefficient, which by (18) is $c\sqrt{\pi}$, must be algebraic. This proves (b). $\square$

This lemma is a special form of the classical asymptotic obstruction to algebraicity, often called Flajolet's criterion; see the discussion in Section 1.4.2 of Bostan–Salvy–Singer [BSS]. The proof above is included to specify the hypotheses and avoid any appeal to a unique dominant complex singularity.

5. Transcendence in every rank at least four

For $m \equiv 0, 1 \pmod 4$, $\alpha_m = m(m-1)/4$ is an integer. If $m \geq 4$, it is positive, so part (a) of the lemma applies directly to (3).

For $m \equiv 2, 3 \pmod 4$, α_m is a half-integer. In these cases $r = \lfloor m/2 \rfloor$ is odd. If $m \geq 4$, necessarily $m \geq 6$ and $r \geq 3$. Equation (13) yields

$$c_m \sqrt{\pi} = A_m \pi^{-(r-1)/2}. \tag{19}$$

Here $(r-1)/2$ is a positive integer and A_m is nonzero algebraic. Since π is transcendental, the right side of (19) is transcendental: if it were algebraic, a positive integral power of π would be algebraic, making π algebraic as well. Part (b) of the lemma contradicts algebraicity of F_m.

These two cases exhaust all $m \geq 4$. Moreover, algebraicity of $F_m(z)$ and of $H_m(t) = F_m(t^2)$ is equivalent. One implication follows by substitution; for the other substitute $t = \sqrt{z}$ and use transitivity of algebraic extensions. Therefore both series are transcendental in every rank at least four.

Although we stated algebraicity over $\mathbf{Q}(z)$, this also gives transcendence over $\mathbf{C}(z)$. For a rational-coefficient formal series, any finite-degree polynomial relation with complex coefficients implies one with rational coefficients. Indeed, fixing the finitely many monomials of the putative relation gives an infinite linear system over $\mathbf{Q}$ with finitely many unknown coefficients. Its row space is finite-dimensional and is spanned by finitely many rows. A nontrivial complex nullspace therefore has a nontrivial rational vector. Thus no change of constant field can evade the obstruction.

6. Algebraicity in the remaining ranks

Ranks zero and one give the first two formulas of (2) directly from (1). For rank two, half the Haar mass of $O(2)$ lies on rotations and half on reflections. A reflection has trace zero. A rotation has trace $2\cos\theta$, with θ uniform. Thus, for $n \geq 1$,

$$a_2(n) = \frac{1}{2}\frac{1}{2\pi}\int_0^{2\pi} (2\cos\theta)^{2n}\, d\theta = \frac{1}{2}\binom{2n}{n}.$$

The integral is the constant term of $(x + x^{-1})^{2n}$. Keeping the constant coefficient $a_2(0) = 1$ gives H_2 in (2).

For rank three, use $SO(3) \cong SU(2)/\{\pm I\}$. Under the double cover, the standard three-dimensional representation is $\mathrm{Sym}^2(\mathbf{C}^2)$. Denote the irreducible spin-j representation by W_j, so W_0 is trivial and W_1 is the standard three-dimensional representation. The Clebsch–Gordan rule is

$$W_j \otimes W_1 \cong W_{j-1} \oplus W_j \oplus W_{j+1} \quad (j \geq 1), \qquad W_0 \otimes W_1 \cong W_1. \tag{20}$$

It follows either from multiplication of the characters $\mathrm{ch}\, W_j = x^{-j} + \cdots + x^j$ or from the usual $SU(2)$ tensor rule.

Consequently the multiplicity of W_0 in $W_1^{\otimes N}$ counts nonnegative Motzkin excursions of length N with no horizontal steps at height zero. Let their generating function be $R(u)$. Let $M(u)$ count ordinary Motzkin excursions, where horizontal steps at height zero are allowed. First-step decomposition gives

$$M = 1 + uM + u^2M^2, \qquad R = \frac{1}{1 - u^2M}. \tag{21}$$

The second identity follows by decomposing an excursion counted by R into a sequence of primitive excursions; each primitive is an up step, an arbitrary Motzkin excursion one level higher, and a down step. Solving the quadratic with $M(0) = 1$ yields

$$M(u) = \frac{1 - u - \sqrt{1 - 2u - 3u^2}}{2u^2}, \qquad R(u) = \frac{1 - \sqrt{(1-3u)/(1+u)}}{2u}. \tag{22}$$

The formula for R in (22) is the binary-quadratic example of Almkvist–Dicks–Formanek [ADF, Example 5.10, p. 207]. Their representation is $\mathrm{Sym}^2(\mathbf{C}^2)$, not a family of defining orthogonal representations in increasing dimension.

The group $O(3)$ is generated by $SO(3)$ and the central element $-I$, which acts on $V^{\otimes N}$ by $(-1)^N$. Therefore its even-degree invariant dimensions equal those for $SO(3)$, and the odd-degree dimensions vanish. Taking the even part $(R(t) + R(-t))/2$ gives exactly H_3 in (2). This proves algebraicity in rank three without identifying a sequence solely from its first few terms.

For polynomial witnesses without radicals, the compressed series satisfy

$$(1-4z)(2F_2-1)^2 - 1 = 0, \qquad 4z(1-z)^2F_3^4 - (1-z)(1+3z)F_3^2 + 1 = 0. \tag{23}$$

Each equation has a unique formal branch with constant term 1: its derivative with respect to the series variable at $(z, F) = (0, 1)$ is nonzero. These equations follow by squaring (2), with the branch fixed before squaring.

7. D-finiteness and the original counterexample

The tca $A = \mathrm{Sym}(\mathrm{Sym}^2)$ is finitely generated and $A_0 = \mathbf{C}$. Its quotient B_m is a cyclic, hence finitely generated, A-module. Section 2 shows that it is bounded by m rows. Sam–Snowden [SS, Theorem 6.1] therefore applies to its exponential Hilbert series

$$E_m(t) = \sum_{d \geq 0} \dim(B_m)_d \frac{t^d}{d!}.$$

This series is D-finite. The formal series $\sum_{d\geq 0} d!t^d$ is D-finite, since its coefficients satisfy $b_{d+1} = (d+1)b_d$. The Hadamard product of these two D-finite series is $H_m(t)$, so H_m is D-finite as well. This is precisely the ordinary-versus-exponential conversion explained in [SS, Section 7.4]. D-finiteness is preserved by algebraic substitution, so $F_m(z) = H_m(\sqrt{z})$ is also D-finite. Rank zero is immediate.

For comparison with the original yes/no question, rank four gives the particularly explicit formula

$$a_4(n) = \frac{C_n^2 + C_n}{2}, \qquad C_n = \frac{1}{n+1}\binom{2n}{n}. \tag{24}$$

Here is a proof of the normalization. Under $\mathrm{Spin}(4) \cong SU(2) \times SU(2)$, the complex standard four-dimensional representation is $W \otimes W$, with $W = \mathbf{C}^2$. Regrouping tensor factors identifies the invariant space in degree $2n$ for $SO(4)$ with $U_n \otimes U_n$, where $U_n = (W^{\otimes 2n})^{SU(2)}$ has dimension C_n. The latter Catalan dimension follows from Schur–Weyl duality: the trivial SL_2 representation occurs in the rectangular shape (n, n), whose hook-length dimension is C_n.

The tensor swap $v \otimes w \mapsto w \otimes v$ preserves the symmetric form obtained as the tensor product of the two alternating forms on W. Its $+1$ eigenspace has dimension three and its -1 eigenspace dimension one, so it has determinant -1. It therefore represents the other component of $O(4)$. On $U_n \otimes U_n$ its action is the swap, and the fixed space has dimension $C_n(C_n + 1)/2$. This proves (24), including $n = 0$. In particular $c_4 = 1/(2\pi)$ also follows from Catalan asymptotics.

The Catalan-square series is already the nonalgebraic example in [SS, Example 7.11]; the rank-four moment sequence is OEIS A246860. Formula (24) is included for comparison, not offered as a new isolated counterexample. The all-rank proof above does not depend on (24).

8. Verification scope and conclusion

The companion standard-library script `verify.py` performs exact integer and rational checks. It compares the hook-length formula with the removable-corner recurrence for Specht dimensions; compares the rank-three invariant multiplicities with the independent Clebsch–Gordan walk recurrence; expands the explicit radicals; tests (24); and checks the arithmetic normalization of the leading constants. It also includes negative controls for the factor of two in rank two and for substituting $SO(4)$ invariants for $O(4)$ invariants.

These computations are diagnostic checks of the formulas and conventions. They are not an all-rank computational certificate. The all-rank result is established by the invariant-theory identification, the geometric asymptotic proof, the radial lemma, and the two exhaustive residue classes in Section 5. Standard foundational inputs are stated where used: the orthogonal fundamental theorems and restriction identity, Schur–Weyl duality, Newton–Puiseux, the transcendence of π, and the cited D-finiteness theorem and closure properties.

Thus the exact algebraicity boundary for the reduced symmetric determinantal quotients in (1) is rank three. The original proposed algebraicity assertion fails starting at rank four, and algebraicity never returns at any larger finite rank.

9. Statement

Fix an integer $m \geq 1$, the defining representation $V = \mathbf{C}^m$ of $O(m, \mathbf{C})$, and a nonzero finite-dimensional algebraic representation W. Define

$$b_W(N) = \dim \mathrm{Hom}_{O(m)}(W, V^{\otimes N}), \qquad P_W(t) = \sum_{N \geq 0} b_W(N) t^N.$$

These are multiplicities, not dimensions of whole isotypic components. Multiplying by $\dim W$ gives the latter only when W is irreducible.

Theorem. Every P_W is D-finite, and P_W is algebraic over $\mathbf{C}(t)$ if and only if $m \leq 3$. Equivalently the same statement holds over $\mathbf{Q}(t)$, because its coefficients are integers. The zero representation is excluded: its series is zero in every rank.

This enlarges the scalar-invariant statement of Section 1 to all fixed covariants. It is not a theorem about arbitrary modules over a determinantal tca, virtual differences of representations, or representations varying with tensor degree. Its historical novelty remains subject to the accompanying prior-work comparison; the character-integral method and multiplicity asymptotics are not claimed as new.

10. Higher rank: the obstruction survives every fixed representation

First suppose that the central element $-I$ acts on W as $(-1)^\epsilon$, with $\epsilon \in \{0, 1\}$. This holds for every irreducible W. Character orthogonality on the compact real group $K = O(m, \mathbf{R})$ gives

$$b_W(N) = \int_K \overline{\chi_W(g)} (\mathrm{tr}\, g)^N \, dg.$$

Incompatible parity gives zero, either by replacing g by $-g$ or by comparing the action of $-I$ in the Hom space. In compatible degree $N = 2n + \epsilon$, the two trace maxima are I and $-I$. In the coordinates and metric of Section 3 above, write $D = m(m-1)/2$ and $\alpha = D/2$. At I,

$$\operatorname{tr}(\exp A) = m - \|A\|^2 + O(\|A\|^4), \qquad \overline{\chi_W(\exp A)} = \dim W + O(\|A\|).$$

At $-I$, the character contributes $(-1)^\epsilon$ and the tensor power contributes $(-1)^{2n+\epsilon}$; their product is 1. Thus the two leading contributions add, rather than cancel. The character is bounded on the compact group, so the complement of these two neighborhoods remains exponentially negligible. The same dominated-convergence rescaling as in the scalar proof now yields, for every fixed $m \geq 2$ and fixed W,

$$b_W(2n+\epsilon) \sim (\dim W) m^\epsilon c_m m^{2n} n^{-\alpha}, \qquad c_m = 2^{1-m}(m/2)^\alpha \pi^{-m/2} \prod_{j=1}^{m} \Gamma(j/2).$$

This argument does not require the character to be positive away from the maxima. Smoothness at the maxima and boundedness elsewhere suffice. It proves eventual positivity on the compatible parity, in particular that a nonzero W cannot yield a polynomial series.

Apply the radial obstruction of Section 4 above to $Q_W(z) = \sum_{n \geq 0} b_W(2n+\epsilon) z^n$. For $m \equiv 0, 1 \pmod 4$, $m \geq 4$, the exponent α is a positive integer and a derivative has logarithmic divergence. For $m \equiv 2, 3 \pmod 4$, $m \geq 6$, the corresponding square-root amplitude is the scalar amplitude multiplied by the nonzero integer $(\dim W) m^\epsilon$. It remains transcendental. Hence Q_W, and therefore $P_W(t) = t^\epsilon Q_W(t^2)$, is nonalgebraic.

For arbitrary W, split $W = W_+ \oplus W_-$ into its two central-parity eigenspaces. The even and odd parts of P_W are precisely P_{W_+} and P_{W_-}. Algebraic series are closed under parity extraction. At least one of these representations is nonzero, giving the contradiction just proved. Positivity of genuine representation multiplicities is essential here; no analogous statement is made for virtual representations.

11. All low-rank multiplicity series explicitly

For $m = 1$, $O(1) = \{\pm 1\}$. Its trivial and sign representations have series $1/(1-t^2)$ and $t/(1-t^2)$, respectively. Finite direct sums give all cases.

For $m = 2$, the irreducibles are the trivial and determinant characters and the two-dimensional representations U_k, $k \geq 1$, with rotation character $x^k + x^{-k}$ and reflection character zero. Put

$$s = \sqrt{1-4t^2}, \qquad q = \frac{1-s}{2t}.$$

Then

$$P_{\mathbf{1}} = \frac{1+s^{-1}}{2}, \quad P_{\det} = \frac{s^{-1}-1}{2}, \quad P_{U_k} = \frac{q^k}{s}.$$

To check the last formula, the character integral is the constant term of $x^{-k}/(1-t(x+x^{-1}))$ on the rotation component, after its half mass cancels the two equal terms of $x^k + x^{-k}$. If A_k denotes this constant term, it satisfies $A_k = t(A_{k-1} + A_{k+1})$ for $k \geq 1$, and $A_0 = 1 + 2tA_1$. The solution with A_k of valuation at least k is $A_k = q^k/s$. For the one-dimensional characters the reflection component contributes $\pm 1/2$ only in degree zero, proving the first two formulas.

For $m = 3$, $O(3) = SO(3) \times \{\pm I\}$. Its irreducibles are $W_{j,\epsilon}$, $j \geq 0$, $\epsilon \in \{0, 1\}$, whose restriction to $SO(3)$ is spin j and whose central parity is ϵ. Set

$$q(t) = \frac{1 - t - \sqrt{1 - 2t - 3t^2}}{2t}, \qquad R(t) = \frac{1}{1 - tq(t)}.$$

If $A_j(t)$ is the multiplicity generating function for spin j in powers of spin one, the Clebsch–Gordan rule gives

$$A_0 = 1 + tA_1, \qquad A_j = t(A_{j-1} + A_j + A_{j+1}) \quad (j \geq 1).$$

The solution is $A_j = Rq^j$: the quadratic $tq^2 - (1 - t)q + t = 0$ verifies the bulk equations and $R(1 - tq) = 1$ verifies the boundary. Uniqueness holds coefficient by coefficient, since each degree is determined by the preceding degree and initially only $A_0(0) = 1$ is nonzero. Central-parity extraction therefore gives

$$P_{W_{j,\epsilon}}(t) = \frac{R(t)q(t)^j + (-1)^\epsilon R(-t)q(-t)^j}{2}.$$

Every finite direct sum of these formulas is algebraic. This proves all low-rank cases, not just the scalar representation.

12. Why D-finiteness extends to these covariants

Let $M_W(U) = \mathrm{Hom}_{O(m)}(W, \mathrm{Sym}(U \otimes V))$ as a polynomial functor of the auxiliary space U. Its symmetric sequence has dimension $b_W(N)$ in degree N. It is a module over $B_m(U) = \mathrm{Sym}(U \otimes V)^{O(m)}$.

We verify the finite-generation hypothesis rather than assume it for arbitrary covariants. Every finite-dimensional algebraic $O(m)$ representation is a direct summand of a finite direct sum of tensor powers of V. Indeed the coordinate algebra of $O(m)$ is generated by matrix entries (the inverse matrix is its transpose), each polynomial degree is a quotient of a finite direct sum of tensor powers of V with multiplicity, and every irreducible occurs in its matrix coefficients. Complete reducibility gives the required direct-summand assertion.

It suffices to treat $W = V^{\otimes d}$. By self-duality, a multilinear element of M_W is an invariant contraction of $N + d$ positions, of which d are designated output positions. The orthogonal first fundamental theorem spans these invariants by pairings. In each pairing, every pair joining two input positions factors off as a generator of B_m. What remains pairs output positions with each other or with at most d input positions. Consequently the symmetric-sequence module is generated by its finite-dimensional pieces in input degrees at most d. This is finite generation as a polynomial-functor/tca module, not finite generation of its total vector space. Relations among pairings do not invalidate the spanning assertion. Direct sums and module projections preserve finite generation, so M_W is finitely generated over B_m and over its finitely generated parent tca.

Schur–Weyl decomposition of $\mathrm{Sym}(U \otimes V)$ shows that M_W has no Schur functors with more than m rows. It is bounded. Sam–Snowden, Theorem 6.1, applies to its exponential Hilbert series. Taking the Hadamard product with $\sum N! t^N$ gives the ordinary series P_W, as in Section 7 above. Thus P_W is D-finite for every fixed W and m.

13. Consequences and limitations

The scalar determinantal quotient is the case $W = \mathbf{1}$. The theorem shows that its higher-rank transcendence is shared by every fixed nonzero covariant type, including the determinant character and all fixed tensor symmetries. Passing from scalars to a chosen fixed type does not restore algebraicity. Conversely, in dimensions at most three all these series have the explicit algebraic formulas above.

For $m \geq 4$, no polynomial equation over $\mathbf{C}(t)$ can encode any such nonzero fixed-type multiplicity series, although a linear differential equation with polynomial coefficients exists. A finite polynomial system whose solution component is known to be algebraic cannot produce it either. This does not exclude every implicit polynomial system: underdetermined systems can admit arbitrary nonalgebraic solutions. Nor does the result give an efficient algorithm or a differential-equation order bound.

The compatible-parity asymptotic also gives a dimension ratio: for two fixed nonzero representations of the same central parity,

$$\frac{b_W(2n+\epsilon)}{b_Z(2n+\epsilon)} \longrightarrow \frac{\dim W}{\dim Z}.$$

This is a consequence, not a claimed new limit-ratio principle. Classical tensor-power asymptotic work must be credited for that phenomenon.

14. Prior work, consequences and publication boundary

For our scalar coefficients, Regev's row-divisibility family is exactly

$$a_m(n) = U^{(1)}_{2,m}(2n).$$

His Theorem 3.3 gives $U^{(1)}_{d,m}(N) \sim d^{1-m} S^{(1)}_m(N)$, while (F.4.5.1) gives the full constant for $S^{(1)}_m$. Substitution $d = 2$, $N = 2n$ yields precisely (3)–(4). This is an exact antecedent, not just a nearby strip estimate. His column-divisibility family is different and is not used for this identification.

ADF's Theorem 2.1 gives the noncommutative Molien–Weyl integral for the same tensor-algebra invariant object. Their Section 4.2 proves algebraicity for SL(2) tensor-algebra invariants by one-variable residue calculus. Their Example 5.10 gives (22), and taking its even part gives our rank-three scalar series. The printed coefficient 93 in degree eight is corrected to 91 in the author's erratum. Their binary-form degree parameter is not our orthogonal dimension, so their all-binary-form result is not an all-dimension orthogonal classification.

Grabiner–Magyar [GM, Sections 2–3] develops tensor-multiplicity and chamber-walk generating functions. Tate–Zelditch [TZ, Section 0.2, Theorem 8, citing Biane] records fixed-type dimension-proportional tensor-power asymptotics in the connected semisimple setting. We do not claim the dimension-ratio phenomenon, character integration, or weighted Laplace method as new. The present proof spells out the full disconnected orthogonal group's central-parity contribution.

The inspected sources do not state the exact all-fixed-O(m)-type algebraicity classification given here. This bounded comparison is not an exhaustive priority determination. Cohen–Regev (1988), DOI 10.1137/0519085, was unavailable and is not represented as inspected; Regev (1981) already

settles the attribution of the scalar asymptotic. The detailed theorem-to-theorem audit and source hashes accompany the paper.

The classification has a uniform consequence: choosing a different genuine fixed representation type never repairs higher-dimensional nonalgebraicity. Its exclusion of virtual representations is essential. For example, in dimension four the trivial and determinant series have respective coefficients $(C_n^2 + C_n)/2$ and $(C_n^2 - C_n)/2$ in degree $2n$. Their difference is the algebraic Catalan series. The second formula follows from the antisymmetric, rather than symmetric, subspace of $U_n \otimes U_n$ in Section 7. Thus a virtual difference can cancel the obstruction even though neither genuine type is algebraic. This illustration is not claimed as a new Catalan identity.

The result separates algebraic generating-function descriptions from D-finite ones. It supplies no effective differential-equation order bound, complexity guarantee, positive-characteristic result, varying-type asymptotic, or theorem for arbitrary modules or nonreduced thickenings. External specialist review, independent reproduction and formal verification remain unestablished. Publication and internal AI editorial review do not settle correctness, novelty or priority.

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